Practice Problem Set with Detailed Solutions

Chapter 6: Economic Growth, the Financial System, and Business Cycles

Hubbard, O'Brien, and Serletis (4th Canadian Edition)

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Part A – Conceptual Questions

1. Economic Growth vs. Business-Cycle Expansion

Question: Explain the difference between long-run economic growth and short-run business-cycle expansion. Why is sustained growth primarily determined by productivity rather than labour hours?

Solution:

- **Economic growth** refers to the long-term increase in real GDP per capita — that is, sustained increases in the productive capacity of the economy.
- **Business-cycle expansion** refers to short-term increases in real GDP as the economy recovers from a downturn or moves toward full employment.
- Sustained growth depends on productivity (output per worker) because there is a limit to how much total labour hours can increase — workers can only work so many hours. Productivity gains, driven by technological progress and capital accumulation, allow output to grow even when labour hours remain constant.

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2. Productivity and Institutions

Question: List three key factors that determine labour productivity and discuss how the financial system contributes to each.

Solution:

1. **Physical capital:** Machinery, equipment, and buildings that enhance worker efficiency. Financial markets supply funds for firms to invest in new capital.
2. **Human capital:** Skills and education that improve worker quality. Financial systems enable student loans and corporate training investments.

3. **Technological progress:** Innovation and improved production methods. Banks, stock markets, and venture capital provide financing for R&D and innovation.

The financial system channels savings into productive investment, ensuring resources are used efficiently to enhance productivity.

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3. Functions of the Financial System

Question: Describe three key roles of the financial system in promoting economic growth.

Solution:

- **Risk sharing:** Allows savers to spread investments across various assets, reducing risk exposure.
- **Liquidity:** Enables conversion of assets into cash easily, encouraging participation in investment markets.
- **Information:** Prices of financial assets reflect information about firms and projects, directing funds to their most productive uses.

These functions facilitate efficient allocation of savings into investments, supporting capital accumulation and growth.

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4. Crowding Out Effect

Question: Explain how a government budget deficit affects the market for loanable funds, the equilibrium interest rate, and private investment.

Solution:

- A government deficit increases demand for loanable funds (shifts demand rightward).
- As a result, the **real interest rate rises**.
- Higher interest rates discourage private firms from borrowing to invest, leading to **crowding out**.

Step-by-step explanation:

1. In equilibrium, savings = investment (loanable funds market).

2. When government borrows more, total demand for funds increases.
3. Supply of funds is relatively fixed in the short run.
4. The increased demand drives up interest rates, which makes private borrowing more expensive.

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5. Phases of the Business Cycle

Question: List the four phases of the business cycle and describe the typical movements of GDP, unemployment, and inflation during each phase.

Solution:

1. **Expansion:** Output, income, and employment rise; unemployment falls; inflation begins to rise.
2. **Peak:** Output reaches its temporary maximum; inflation is often at its highest.
3. **Recession:** Output and employment fall; unemployment rises; inflation declines.
4. **Trough:** Economic activity bottoms out; inflation is low; recovery begins thereafter.

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Part B – Quantitative Problems with Step-by-Step Solutions

6. Growth Rate Calculations

Question: Real GDP in Canada increased from $1.8 trillion in 2015 to $2.2 trillion in 2020. (a) Compute the average annual growth rate. (b) Using the Rule of 70, estimate how long it would take for GDP to double at this rate.

Solution:

Step 1: Calculate growth rate

$$g = \left(\frac{2.2}{1.8}\right)^{1/5} - 1$$

$$g = (1.2222)^{0.2} - 1 = 0.041 = 4.1\%$$

Step 2: Rule of 70

$$\text{Doubling time} = \frac{70}{g} = \frac{70}{4.1} \approx 17.1 \text{ years}$$

Interpretation: At 4.1% annual growth, real GDP would double in about 17 years.

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7. Per Capita Growth Rate

Question: Country A's real GDP grows by 5% per year, while its population grows by 2% per year. (a) Find the growth rate of real GDP per capita. (b) Interpret your result.

Solution:

$$\text{Real GDP per capita growth} = 5\% - 2\% = 3\%$$

Interpretation: On average, citizens' living standards (output per person) rise by 3% annually.

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8. Market for Loanable Funds

Question: Suppose demand for loanable funds is $D = 800 - 50r$ and supply is $S = 200 + 100r$, where r is the real interest rate (in %). (a) Find equilibrium r and Q. (b) If government borrowing increases by 100 (shifts demand right), find the new equilibrium.

Solution:

Step 1: Equilibrium condition

$$D = S \Rightarrow 800 - 50r = 200 + 100r$$

$$600 = 150r \Rightarrow r = 4\%$$
$$Q = 200 + 100(4) = 600$$

Equilibrium: $r^* = 4\%, Q^* = 600$

Step 2: With government borrowing (+100 demand)

$$D' = 900 - 50r$$

$$900 - 50r = 200 + 100r \Rightarrow 700 = 150r \Rightarrow r' = 4.67\%$$
$$Q' = 200 + 100(4.67) = 667$$

Interpretation: Higher government borrowing raises interest rates and reduces private investment due to crowding out.

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9. Productivity and Technology – Solow Type Example

Question: Output is given by $Y = AK^{0.3}L^{0.7}$. Initially, $A = 1$, $K = 100$, $L = 100$. (a) Find initial output. (b) If A increases by 10%, find new output and percentage increase.

Solution:

Step 1: Compute initial output

$$Y_0 = 1 \times 100^{0.3} \times 100^{0.7} = 1 \times 100 = 100$$

Step 2: After 10% increase in A

$$A_1 = 1.1, \quad Y_1 = 1.1 \times 100^{0.3} \times 100^{0.7} = 110$$

$$\%\Delta Y = \frac{110 - 100}{100} \times 100 = 10\%$$

Interpretation: A 10% productivity improvement raises output by exactly 10%, holding capital and labour constant.

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10. Business Cycle Data Interpretation

Question: Use the following table of real GDP growth rates to identify recession and recovery periods:

Year	2018	2019	2020	2021	2022
Growth Rate (%)	2.6	1.8	-4.5	4.1	2.0

(a) Which year represents recession and which recovery? (b) How would unemployment and inflation behave in these years?

Solution:

- (a) 2020 shows a recession (negative growth). 2021 marks a recovery (positive rebound).
- (b) During 2020: output falls, unemployment rises, and inflation falls. During 2021: economic recovery reduces unemployment and increases inflationary pressure.

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Part C – Applied Discussion

11. Policies to Promote Growth

Question: Suggest two policies that could increase long-run economic growth in Canada and explain their short-run effects on the business cycle.

Solution:

- **Investment incentives:** Tax credits for R&D or corporate investment raise capital formation and productivity. In the short run, such policies boost aggregate demand, reducing unemployment.
- **Education and training programs:** Enhance human capital, improving long-term productivity. In the short run, they stimulate spending and employment in the education sector.

Both policies increase potential GDP and smooth business-cycle fluctuations.

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12. Savings and Investment Equality

Question: Explain how the equality of savings and investment ensures efficient resource use in the economy.

Solution:

Step 1: National income identity

$$Y = C + I + G$$

Subtract C and G from both sides:

$$Y - C - G = I$$

By definition, $S = Y - C - G$, so:

$$S = I$$

Step 2: Interpretation

- Savings provide the funds for investment.
- When savings = investment, all income not consumed is used for productive purposes.
- If investment falls short of savings, output is underutilized — leading to recessions.

Hence, the equality maintains balance between current production and future growth.

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End of Problem Set